

CO₂ Emissions and Income Dynamics: What Does the Global Evidence Tell Us?

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Abstract This paper analyzes the co-evolution of two major determinants of social welfare, namely, income and carbon emissions. In particular, by using a distribution dynamics approach based on Markov chains, we investigate the shape and behavior of the joint distribution of per-capita income and carbon dioxide emissions. We arrive at several interesting conclusions, especially in the context of international negotiations on climate change. First, evidence does not support theoretical models predicting the existence of a poverty-environment trap. Specifically, in the long-run two main groups of countries will emerge: poor versus polluting countries. Second, the typical development path leads initially to high emission levels and, subsequently, to high income. Third, the convergence process towards the stationary distribution is very slow. Finally, for carbon emissions, whenever it is observed, the environmental Kuznets curve seems to be only a transitory phenomenon.

Keywords CO₂ emissions · Convergence · Markov chains · Multiple equilibria

JEL Classification C14 · O11 · O44 · Q53

1 Introduction

Air pollution, the release of chemicals and particulates into the atmosphere, is widely recognized to affect health and social welfare negatively. The primary effects of air pollution

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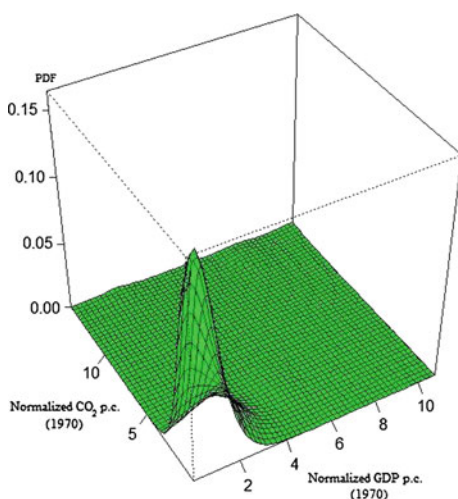
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Fig. 1 Probability density function (1970)



include smog, acid rain, global warming and other damages to the ozone layer. According to the latest estimates by the International Energy Agency (IEA 2011), energy-related carbon dioxide (CO₂) emissions in 2010 were the highest in history. For this reason, public intervention and international coordination efforts are developing to combat air pollution and thus increase environmental quality.¹ In this context, the income–pollution relationship has captured the attention of policymakers, theorists and empirical researchers alike, especially in the last two decades. Thus far, empirics have failed to reach conclusive results. By considering the joint dynamics of CO₂ emissions and income, this paper fills a gap in the existing literature, where a satisfactory analysis of the dynamic interaction between income and air pollution is still missing. Throughout this article, the terms “pollution” and “air pollution” always refer to CO₂ emissions.

Considering a sample of 126 countries, Figs. 1 and 2 below present the estimated joint distributions of per capita gross domestic product (GDP) and per capita CO₂ emissions in 1970 and 2006, respectively.² Data come from the World Bank’s World Development Indicators (WDI 2011), and they have been normalized with respect to the sample average. By comparing the initial and final distributions, we observe the emergence of a second peak characterized by higher income and pollution levels.³ This suggests that with time a long-run multimodal distribution might emerge, which reveals the existence of *club convergence* among countries. This evidence further motivates our study.⁴ Note that we refer to multiple

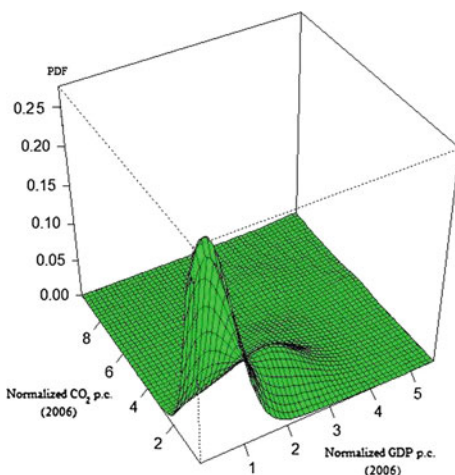
¹ For example, see US President’s Obama’s speech at the [State of the Union \(2010\)](#) and United Nations Framework Convention on Climate Change (known as the “Kyoto Protocol”) (1997).

² These distributions are estimated by using two smooth kernel functions centered over each observation and are based on a nonparametric method provided by [Bowman and Azzalini \(1997\)](#). See “Appendix A” for details.

³ While in Fig. 1 we can identify only one peak corresponding to a GDP p.c. lower than the world average, in Fig. 2 a second peak emerges. This peak corresponds to countries with a GDP level at least two times higher than the world average. These countries are also characterized by emission levels that are about three times higher than those corresponding to the first peak. See also Fig. 7 in the “Appendix”, where the solid line represents the contour plot of Fig. 1, and the dashed line represents the contour plot of Fig. 2.

⁴ Notice that a single peaked distribution, even in 2006, would not preclude the existence of underlying dynamics, leading to multiple peaks. For this reason, the use of a Markov-chain approach to investigate these latent dynamics is justified.

Fig. 2 Probability density function (2006)



equilibria in terms of long-run levels of income and air pollution and not in the sense of multiple long-run distributions.

In this work, we pay attention to CO₂ emissions for a number of reasons. First, CO₂ emissions are responsible for global warming, as they constitute more than 75 % of the greenhouse gas emissions (World Bank 2010). This is the reason why they are the main objective of international negotiations on climate change (e.g., the Kyoto Protocol).⁵ Second, CO₂ emissions are strongly related to environmental quality, as they are positively linked with the Ecological Footprint index (correlation 0.815), which shows how much biologically productive land and water a population requires to support current levels of consumption and waste production, using prevailing technology.⁶ Finally, CO₂ is the most commonly used environmental quality indicator in the environment-income literature, and our results would thus be directly comparable with previous literature (see, e.g., Narayan and Narayan 2010; Acaravci and Ozturk 2010).⁷

We jointly analyze the shape and behavior of the distribution of carbon emissions and income as well as the mobility of countries within this distribution. To do so, we employ nonparametric techniques, namely, stochastic kernels and Markov chains, because cross-section and panel-data econometric techniques, which study the behavior of the “average” economy, are not suitable. Our study makes a contribution in the environment-income nexus in several important ways. First, by taking an agnostic view on *equilibrium multiplicity* versus *uniqueness*, our methodology allows us to test the validity of theoretical models predicting environment and/or poverty traps (see, among others, John and Pecchenino 1994; Xepapadeas 1997; Mariani et al. 2010). Second, our framework enables us to reveal divergence/convergence trends using a world sample. Indeed, a Markov chain analysis can tell us if countries are converging towards a unique long-run equilibrium, or if they are converging towards

⁵ Until recently, carbon dioxide emissions had been perceived solely as a global issue. However, recent studies have shown they also have a local impact, harming mostly the health of urban populations (see Dudek et al. 2003; Jacobson 2008; Garg 2011).

⁶ Global Footprint Network (2011), United Nations Framework Convention on Climate Change (1997).

⁷ More recently, a number of sustainability indices have been constructed, such as the Environmental Performance Index (Esty et al. 2008) and the Sustainable Society Index (Van de Kerk and Manuel 2008). Such indices include multiple environmental and social indicators combined together. However, this implies substitutability between environmental and social goals. We believe that these two dimensions should be treated separately, as more society does not compensate for less environment, and vice versa (O'Neill 2011).

multiple equilibria. Third, we introduce a novel definition of the state-space, which includes income and air pollution.⁸ Finally, we conduct inferences on the relationship between the development stage and pollution dynamics of countries, to draw policy implications.

We arrive at several conclusions. First, our results show a strong long-run positive relationship between income and CO₂ emissions. We predict that at the end of the dynamic process, the world will be divided into two main groups: low-income/low-pollution countries and high-income/high-pollution countries. Steady state predictions reveal that around 63 % of countries will be characterized by high pollution if our consumption-production behavior remains unaltered. In other words, our results do not validate models in which countries can be caught in a poverty-environment trap. Second, the most likely development path of a low-income/low-pollution country, if it manages to escape from the poverty trap, is that it first increases its pollution level and then enjoys higher income. Third, regarding transitional dynamics, our estimates show that countries tend to be strongly persistent with respect to their income-pollution status, that is, they are very likely to remain in the current state in the future. Besides, the convergence process towards the steady state is very slow. However, given the fact that current emissions are already extremely high, this finding gives citizens and policy makers a limited window of time to take action and return to a sustainable development path.⁹ This requires more drastic changes in economic structure than those that have been achieved thus far. Finally, we do not obtain support for the environmental Kuznets curve (EKC) hypothesis for CO₂ emissions.

The rest of the article is structured as follows. In Sect. 2, we relate our contribution to the existing literature. In Sect. 3, we present the dataset, and in Sect. 4 we outline the empirical methodology. In Sect. 5, we estimate the transition probabilities between states in the income-pollution space and the stationary distribution of countries. We also compute some mobility indices and test the robustness of our results. In Sect. 6, we provide concluding remarks and policy implications. The Appendices include details of the empirical analysis.

2 Related Literature

Our analysis is related to both theoretical literature on equilibrium multiplicity and empirical literature on convergence towards one or more equilibria. In particular, using a distribution dynamics approach, one can investigate the long-run behavior of a stochastic process. If the ergodic distribution shows a tendency towards a point mass, then we can conclude that the stochastic process is converging towards a unique stationary equilibrium. On the contrary, if the stochastic process shows tendencies towards limits that have different properties—for instance, a multi-peaked distribution—then we can conclude that countries exhibit a *club convergence* behavior. In other words, Markov chains can reveal the existence of multiple equilibria as well as the mass of countries converging towards each equilibrium. Below, we discuss the main contributions to these two strands of literature.

2.1 Theoretical Literature

The theoretical framework of our paper comprises the literature on poverty and environmental traps, i.e., situations in which countries remain persistently trapped in a state of

⁸ See Quah (1993) and Fiaschi and Lavezzi (2003, 2007) for similar studies on income levels and growth rates.

⁹ According to the 2010 World Bank's report, by 2030, OECD countries must reduce their power-generation emissions by 70 % and their carbon energy intensity by approximately 38 % compared to 2007 levels.

low income and/or low environmental quality.¹⁰ According to this literature, a poverty trap may arise when there are strong externalities and strategic complementarities affecting an economic system. An early work is by John and Pechennino (1994), who study the relationship between growth and environmental quality in an overlapping generations (OLG) model where short-lived agents make decisions on environment and capital accumulation with long-lasting impacts on factor productivity and the environment. Their model exhibits a case where economies with sufficient capital and environmental quality experience growth and improving environmental quality, while economies with worse initial conditions converge towards a low-capital, low-environmental quality equilibrium, despite increasing returns to production. Xepapadeas (1997) analyzes an endogenous growth model with productive and abatement capital as well as increasing returns due to knowledge spillovers in production and pollution abatement. He shows that countries with environmental concerns can be trapped in a low-growth, high-pollution equilibrium because of insufficient knowledge of pollution abatement. On the contrary, Mäler et al. (2003) analyze a Solow-type model with emissions flow as a production input and nonlinear feedbacks in the environmental system. They demonstrate the possibility of locally stable equilibria with low-capital, low-pollution stocks and high-capital, high-pollution stocks; the steady state at which the economy arrives depends on the degree of pollution decay and the size of the emissions flow.

Ikefuji and Horii (2007) investigate the link between poverty and environmental degradation in an OLG model with agent heterogeneity, environmental externalities, human capital accumulation and credit constraints. They assume that environmental quality influences labor productivity and wealth dynamics through agents' ability, while wealth distribution determines if agents rely on primitive technology that reduces environmental quality. This interaction may lead an economy to a *poverty-environment trap*, where lower environmental quality lowers income and the education level, which, in turn, accelerates environmental degradation. Mariani et al. (2010) develop an OLG model with human and physical capital and two-sided feedback between life expectancy and environmental quality. They show that countries beginning with sufficiently low environmental quality and human capital will be caught in a trap of low life expectancy, low environmental quality and low human capital. D'Alessandro et al. (2010) analyze a model with perfect complementarities between capital and energy, two perfectly substitutable energy sources (fossil fuels and alternative energy) and harnessed energy given by past accumulation of alternative energy-sources capacity (AESC). They show that economies will be trapped in a low-income, low-environmental quality equilibrium if the share and productivity of investments in AESC are too low to increase energy availability and the energy-harnessing capacity is small.

Naimzada and Sodini (2010) study the dynamics of an OLG model in which agents make their consumption decisions without taking into account the existence of strategic complementarities. The authors show the possibility of a high-income, low-environment steady state when total factor productivity (TFP) increases over a threshold value and the elasticity of substitution between capital and labor is sufficiently low. Finally, Varvarigos (2010) studies a model where: (a) longevity is positively affected by public-health spending and negatively influenced by pollution, (b) environmental degradation is positively influenced by pollution due to production and is mitigated by public-environment expenditure. He proves that low income-low pollution equilibria are possible depending on the elasticities of: a) environmental damage with regard to pollution; b) environmental improvements with respect to abatement policy; c) output with respect to capital; and d) output with respect to the average

¹⁰ In these models, pollution comes from production and/or consumption activities. Therefore, since approximately 80 % of carbon dioxide is generated by the energy sector—see Halsnæs and Garg (2011)—there is an immediate link between our analysis and the literature we consider.

capital-labor ratio. The likelihood of traps is also a function of the cleanliness of production technology, TFP and the initial conditions.

With respect to this literature, our empirical evidence confirms the existence of multiple steady states. Specifically, two separate traps emerge: a poverty trap (low income and low pollution) and an environment trap (high income and high pollution), which is in line with only part of the theoretical literature outlined above (Mäler et al. 2003; Naimzada and Sodini 2010; Varvarigos 2010).

2.2 Empirical Literature

Our work mainly contributes to the literature on income-CO₂ convergence. In a recent study, Brock and Taylor (2010) employ the Green Solow model as an alternative framework and present robust evidence of convergence in CO₂ levels among 173 countries examined, using standard panel techniques. Similarly, using both cross-country and time-series analyses, Strazicich and List (2003) find evidence of convergence in CO₂ levels among 21 industrial countries. Other studies, such as Romero-Avila (2008) and Westerlund and Basher (2008), provide evidence in favor of CO₂ convergence. By contrast, Barassi et al. (2008) show that OECD countries did not converge in terms of per capita CO₂ emissions during the period 1950–2002. In the same spirit, using non-parametric techniques, Nguyen-Van (2005) finds evidence of convergence only for 26 industrial countries. Therefore, this empirical literature is far from being conclusive.

Markov chains have been used to separately study income and environment dynamics. For instance, considering 118 countries observed for the period 1962–1985, Quah (1993) uncovers a tendency towards a world divided into rich and poor countries, where escaping from the poverty trap is a low-probability event. Fiaschi and Lavezzi (2003, 2007), considering approximately 120 countries for the periods 1960–1989 and 1950–1998, respectively, show that GDP growth first declines, then accelerates and eventually falls as per-capita income rises, following a nonlinear path.¹¹ At the same time, in a study related to our work, Aldy (2006) shows that a Markov-chain analysis does not provide convincing evidence on future emissions convergence. The author finds evidence of convergence among 23 OECD countries, whereas emissions appear to be diverging for a global sample composed of 88 countries for 1960–2000. Recently, Ordás Criado and Grether (2011a), employing, among others, Markov-chain analysis for 166 countries over 1960–2002, find stronger divergence and larger CO₂ per-capita emissions for the world in the long run compared to 2000. They also estimate double median emissions and stable cross-country dispersion during the first 50 years of the transition. Our findings complete the picture initialized by these studies, considering the interaction between income and carbon dioxide emissions, using Markov-chain analysis.

Finally, our approach also sheds some light on the validity of the so-called EKC hypothesis for CO₂ emissions. In general, the EKC hypothesis postulates an inverted U-shaped relationship between pollution and income. In a recent study, Narayan and Narayan (2010) examine a panel of 43 developing countries using panel cointegration to overcome econometric pitfalls; they conclude that CO₂ emissions fall as income rises only in Middle Eastern and South Asian countries. In line with our findings, employing non-parametric methods, Azomahou et al. (2006) and Bertinelli and Strobl (2005) estimate a positive, linear relationship between income and CO₂ emissions. Following Stern (2004), we can say that local pollutants are more likely than global pollutants to display an EKC relationship. That is, the existence of the EKC

¹¹ Markov-chain analysis has also been used to study regional convergence, i.e. Neven and Gouyette (1995), Lopez-Bazo et al. (1999) and Le Gallo (2004).

strongly depends on the nature of the pollutants. For instance, [Ordás Criado et al. \(2011b\)](#), applying semi-parametric and non-parametric estimation for 25 European countries during 1980–2005, show that NO_x and SO_x emissions tend to stabilize along the development path.

3 Data

Our sample consists of cross-country panel data for income and air pollution. As Markov-chain analysis requires long time series together with a large number of cross-sections, we employ per capita GDP (in US\$, at 2005 constant prices) and CO₂ emissions (metric tons per capita), originating from Penn World Tables ([Heston et al. 2009](#)) and World Bank's WDI database ([2011](#)), respectively. CO₂ emissions are originally extracted by the Carbon Dioxide Information Analysis Center, Environmental Sciences Division, USA. This carbon dioxide emissions variable is considered anthropogenic because such emissions are produced from the burning of fossil fuels and the manufacture of cement. They include carbon dioxide produced during consumption of solids, liquids, and gas fuels as well as gas flaring.

Applying two types of panel unit root tests, we first examine the stationarity of our data series in order to verify the existence of markovian dynamics. We assume two alternative unit root processes: a common and an individual one. The tests are by [Levin et al. \(2002\)](#) and by [Breitung \(2000\)](#) for the first type and [Im et al. \(2003\)](#) and [Maddala and Wu \(1999\)](#) for the second one. Both variables are found to be I(1), meaning that the null hypothesis of unit root in levels cannot be rejected, while the null hypothesis of a unit root in first differences can be rejected. In addition, we test for the existence of a structural long-run relationship, i.e. cointegration between our variables. Cointegration occurs when the linear combination of I(1) variables is stationary, implying that deviations of one variable from the path prescribed by the cointegrating relationship are transitory. Applying Pedroni's ([1995, 1997, 1999](#))¹² residual based tests, our results reveal that the null hypothesis of no cointegration is rejected in all cases and thus a Markov chain approach is justified by the data.¹³

Following the literature on Markov chains, we apply mean normalization relative to the sample average to both per capita GDP and CO₂ emissions. As in [Fiaschi and Lavezzi \(2003\)](#), the normalization of per-capita income is justified by the existence of a world trend of technological progress with spillovers to all countries. Regarding the air pollution variable, there is evidence in favor of improvements in environmental quality in some developing countries due to technological transfer from developed ones ([Dasgupta et al. 2002; Stern 2004](#)).

Our main sample includes 126 countries for the 1970–2006 period. A second sample, used as a robustness check, contains 95 countries for 1960–2006.¹⁴ The main sample contains about 200 observations more than the 1960–2006 one and allows us to generalize our conclusions. We observe large disparities with respect to income and pollution, both across countries and over time.¹⁵ Specifically, the income levels between the richest and poorest countries differ by a factor of 220. Additionally, the variance increases by about 50 % in the final year compared to the initial one. By examining the evolution of income, we observe that the poorest quartile of countries increased their per capita GDP by approximately 50 %, while the remaining ones doubled their incomes. Regarding CO₂ emissions, the standard deviation is quite high but decreases considerably over time (approximately 40 %). Finally,

¹² He presents seven tests that can be grouped into two types according to the way in which information is combined. The first type of statistic is based on pooling along the within-dimension of the panel; the second is based on pooling along the between-dimension (see [Pedroni 1999](#)).

¹³ The results for both unit root and cointegration tests are available upon request from the authors.

¹⁴ "Appendix B" provides the country lists for both samples.

¹⁵ See Table 10 in the Appendix for the summary statistics.

Table 1 State-space

Income\CO ₂ emissions	Low < 0.2 μ CO ₂	Middle [0.2 μ CO ₂ , 0.5 μ CO ₂)	High $\geq$ 0.5 μ CO ₂
Low < 0.4 μ GDP	S ₁	S ₂	S ₃
Middle [0.4 μ GDP, 1 μ GDP)	S ₄	S ₅	S ₆
High $\geq$ 1 μ GDP	S ₇	S ₈	S ₉

observing the dynamics of emissions, we note that the least polluting countries increased their emissions by approximately 60 %, while the most polluting ones increased emissions by 20 %. Overall, these findings are confirmed by non-parametric tests that show that the cross-country distributions for both income and air pollution are far from normal.

4 Methodology

4.1 Definition of the State-Space

States of the world are determined jointly by income and pollution levels. By using the contour plot representation of Fig. 1 (Sect. 1), that is, the [Bowman and Azzalini \(1997\)](#) technique described in “Appendix A”, we divide the data into three GDP classes and three CO₂ classes (see “Appendix C” for the graphical splitting). The benchmark state-space is given in Table 1 below. The thresholds that define the country classes are 40%, 100 % of the sample average for per capita GDP and 20 %, 50 % of the sample average for per capita CO₂ emissions.

According to our classification, 40 % of countries lie in the low-GDP class, 30 % in the middle-GDP class and the remaining 30 % in the high-GDP class. This selection of intervals provides us with roughly equally sized classes, as in [Quah \(1993\)](#). Notice that [Quah \(1993\)](#) chooses the state-space exogenously, whereas our state-space definition is at least partially endogenous (using a graphical tool increases the scholar’s role). Regarding CO₂ emissions, 47 % of the countries have low pollution (almost all low-GDP countries), 34 % of countries have high pollution (26 % of these come from high-income countries and another 7 % from middle-income countries), and only 19 % of the sample is characterized by the middle-pollution level (mainly middle-income countries).¹⁶ The threshold value dividing the middle and high-pollution classes corresponds to the sustainable threshold discussed in the Introduction, that is, high pollution also means “unsustainable” pollution.

Our state-space is consistent with the existence of a strong positive correlation between per-capita GDP and CO₂ emissions.¹⁷ Fortunately, Markov chains enable a simple, complete probabilistic description of the system. However, the existence of this strong, positive correlation between per-capita GDP and air pollution implies that most sample-splitting techniques result in no observation in some states of the world (low income/high pollution and high income/low pollution). In this case, we cannot determine the *ergodic distribution*, given the indeterminacy of some transition probabilities.

¹⁶ Some researchers argue that the EKC hypothesis may be due to the reallocation of polluting industries from developed to developing countries ([Stern 2004](#); [Arrow et al. 1995](#); [Suri and Chapman 1998](#); [Copeland and Taylor 2004](#)). However, such a phenomenon stresses the importance of our results even further.

¹⁷ A question that might arise at this point is whether we take into account technological progress and policy initiatives that occurred in the more recent sample years. This is due to the fact that important international environmental agreements and abatement-technology improvements have recently occurred. However, the correlation between per capita GDP and CO₂ emissions remains strongly positive throughout the sample period (0.815 and 0.848 in 1970 and 2006, respectively).

Table 2 State-space as a robust check

Income\CO ₂ emissions	Low < 0.1μCO ₂	Middle [0.1μCO ₂ , 0.5μCO ₂)	High ≥ 0.5μCO ₂
Low < 0.3μGDP	S ₁	S ₂	S ₃
Middle [0.3μGDP, 0.85μGDP)	S ₄	S ₅	S ₆
High ≥ 0.85μGDP	S ₇	S ₈	S ₉

Because the results of a Markov-chain analysis may change significantly when we change the definition of the states, we test the validity of our state-space in several ways.¹⁸ First, and most importantly, we use a stochastic kernel approach to corroborate our transition probabilities. In this way, we can estimate the CO₂ dynamics for each GDP class in a continuous space (see [Fiaschi and Lavezzi 2003, 2007](#)). Second, we change the definition of the states and use a partially different dataset (i.e., less countries and a longer time span) that allow us to verify the robustness of our results. By using the same technique used for Table 1, we split the distribution of countries in 1960, obtaining the state-space described in Table 2. Third, by using a recursive partitioning technique provided by [Breiman et al. \(1984\)](#), we find two critical values for GDP equal to 0.2942 and 0.7766, with 33 % of countries in the low-income class, 29 % of countries in the middle-income class and the remaining 38 % of countries in the high-income class. Adopting the same methodology for CO₂ emissions, we compute two threshold values at 0.1435 and 0.5519. This technique leads to the following division: 44 % of countries are in the low-pollution class, 27 % in the middle-pollution class and the remaining 33 % in the high-pollution class. Note that this division is consistent with Tables 1 and 2. However, we preferred to use a graphical technique because recursive partitioning does not take into account the joint distribution, that is, the GDP and CO₂ thresholds are selected independently.¹⁹

4.2 Estimating the Transition Matrix

We model the income-pollution dynamics as a discrete-time stochastic process, in which the evolution of each country follows a Markov-chain stochastic process with a bivariate state-space. In general, a stochastic process is described by a *stochastic matrix* containing the *transition probabilities* from one state to another.²⁰ If the probability of moving from state i to state j in one period (p_{ij}) depends only on the current state i , the stochastic process is called a Markov chain. We assume that p_{ij} does not change over time, that is, we assume a time-invariant Markov chain. This assumption will be tested later. Therefore, the interpretation of our results will be conditioned on the *ceteris paribus* assumption. Given a certain state-space S , we have that $p_{ij} \in [0, 1]$ and $\sum_{j=1}^S p_{ij} = 1$, $\forall i, j \in S$.

In this framework, following [Norris \(1997\)](#) and [Fiaschi and Lavezzi \(2003\)](#), a maximum likelihood method is used to estimate transition probabilities from one state to another. In particular, we estimate:

¹⁸ [Johnson \(2005\)](#) and [Bulli \(2001\)](#) argue that the process of discretizing the state-space of a continuous variable is necessarily arbitrary and can alter the probabilistic properties of the data.

¹⁹ We can also use cluster analysis, but the main problem with clusters is that there are only a few rich countries, and they constitute a separate group, as is the case with transition economies. This means that we would end up with a state-space partition characterized by the absence of observations in many states.

²⁰ A key property of a stochastic matrix is that it has a *principal left eigenvector* corresponding to its largest eigenvalue, which is 1 ([Norris 1997](#)).

$$\hat{p}_{ij} = \frac{N_{ij}}{N_i}$$

where, N_i is the number of countries in state i at the beginning of the transition period, and N_{ij} is the number of countries that moved from state i to state j during the transition period. Now, from the Ergodic theorem of Markov chains, we know that: if $\mathbf{P}$ is the transition matrix of a regular Markov chain (aperiodic and irreducible), and if $\mathbf{P}^T$ approaches a unique limiting matrix, then the process will reach a steady state distribution $p(T) = p(T + k) = p(0)\mathbf{P}^T$, with $k = 1, 2, \dots$

The stationary distribution is also known as ergodic distribution, and it does not depend on the shape of the initial distribution, $p(0)$. The ergodic distribution captures the final distribution of countries with respect to pollution and income if the structure of the stochastic process is stable. In our case, the shape of the ergodic distribution provides a test of theories on poverty and environmental traps, outlined in Sect. 2.1 of the paper.

4.3 Testing the Time-Invariance Hypothesis

To validate the hypothesis concerning the stationarity of the transition matrix, we use the likelihood ratio test suggested by Anderson and Goodman (1957). The authors give a test for the null hypothesis that the transition matrix is stationary over time against the alternative that transition probabilities are time-variant. That is, we test $H_0 : p_{ij}(t) = p_{ij}(t = 1, \dots, T), \forall i, j \in S$. As in Lopez-Bazo et al. (1999), Neven and Gouyette (1995) and Le Gallo (2004), we divide the sample into two subsamples (1970–1987 and 1988–2006) and then compute the following statistic:

$$Q = -2 \log \left[\prod_{t=1}^2 \prod_{i,j=1}^S \left(\frac{\hat{p}_{ij}}{\tilde{p}_{ij}(t)} \right)^{N_{ij}(t)} \right]$$

where, $\tilde{p}_{ij}(t)$ is the transition probability for subsamples $t = 1$ and $t = 2$. As shown by Anderson and Goodman (1957), the Q-statistic is distributed as χ^2 with $S(S - 1)$ degrees of freedom, where S is the number of states.

Notice that our division of the sample has two main advantages. First, the two subsamples are symmetric. Second, 1988 coincides with the “Toronto Conference on the Changing Atmosphere,” where 46 countries set a 20 % reduction target in CO₂ emissions by 2005 (the so-called “Toronto target”).²¹ This means that the sub-sample 1988–2006 captures a possible structural break in the emissions regimes, providing a reliable test for the time-invariance hypothesis.

4.4 Speed of Convergence and Mobility Indices

We estimate the speed of the convergence process by using the concept of *asymptotic half-life*, i.e., the time required to cover half of the distance from the ergodic distribution. The asymptotic half-life is:

²¹ This was the first major international meeting bringing governments and scientists together to discuss action on climate change. This meeting was also fundamental for the establishment of the Intergovernmental Panel on Climate Change (IPCC), an international group of over 300 of the world’s best climate scientists in charge of reviewing and reporting on the latest international scientific evidence, impacts and responses to climate change.

$$h = -\frac{\log 2}{\log |\lambda_2|}$$

where, λ_2 is the second largest eigenvalue of the transition matrix (see [Fiaschi and Lavezzi 2003](#)).

We also calculate some mobility indices that are useful in understanding how frequent the transitions from one state to another are. Following [Bartholomew \(1982\)](#), we initially consider two indices of mobility. These indices are defined over the interval $[0,1]$ and measure the speed of convergence towards the ergodic distribution. Formally, we have:

$$I_1(P) = \frac{1}{N-1} \sum_i \sum_j p_i(T) p_{ij} |i-j|$$

$$I_2(P) = \frac{1}{(N-1)^2} \sum_i \sum_j p_i(T) p_{ij} (i-j)^2$$

As suggested by [Fiaschi and Lavezzi \(2005\)](#), two other indices can be useful to understanding the mobility of countries during the transition process. These two indices are built on the same idea as the previous ones, but they only consider the transition probabilities:

$$I_3(P) = \frac{1}{A} \sum_i \sum_j p_{ij} |i-j|$$

$$I_4(P) = \frac{1}{A^2} \sum_i \sum_j p_{ij} (i-j)^2$$

where, A is a constant parameter used to normalize the indices to 1.

5 Results

5.1 Main Results

The dynamics of countries with respect to relative per-capita income and CO₂ emissions are represented by the transition matrix reported in Table 3. The first column reports the number of observations for every state. For example, state S_7 has only seven observations; however, a steady state exists.²² The main diagonal represents the probabilities that countries remain in the same state after one year. Along the main diagonal, we can identify three 3x3 blocks, representing the transition process within each GDP class.

In the first block, we have the transition process of countries that will remain in the low-income class in the next period. Countries with low-income/low-pollution have a 99.2 % probability to remain poor in the next period. This probability decreases to 96.4 % for countries with middle-level pollution and 93.6 % for countries with high pollution. For poor countries, it is impossible to move directly into the high-GDP class. Countries with low income/high pollution (S_3) have more chances to become cleaner (9 %) than richer (6.4 %). Nonetheless, we note that countries in state S_3 have the highest probability of escaping from poverty, compared to countries in states S_1 and S_2 . Therefore, state S_1 is a candidate for being

²² The ergodic distribution can be characterized by some states with zero mass. This occurs when there exists only one irreducible closed set of positive persistent aperiodic states and the remaining states are transient (see, e.g., [Isaacson and Madsen 1976](#), p.74).

Table 3 Transition matrix for 1970-2006 over the state-space of Table 1

Obs.	States	S ₁	S ₂	S ₃	S ₄	S ₅	S ₆	S ₇	S ₈	S ₉
1,553	S ₁	0.979	0.013	0	0.007	0.001	0	0	0	0
170	S ₂	0.076	0.835	0.053	0	0.035	0	0	0	0
78	S ₃	0	0.090	0.846	0	0	0.064	0	0	0
220	S ₄	0.064	0	0	0.836	0.091	0	0	0.009	0
519	S ₅	0	0.012	0	0.023	0.913	0.040	0	0.012	0
486	S ₆	0	0	0.010	0	0.029	0.916	0	0	0.045
7	S ₇	0	0	0	0.143	0	0	0.571	0.286	0
60	S ₈	0	0	0	0	0.067	0.017	0.017	0.783	0.117
1317	S ₉	0	0	0	0	0.008	0.015	0	0.008	0.983
Ergodic dist.		0.201	0.035	0.022	0.023	0.099	0.149	0.00	0.008	0.462

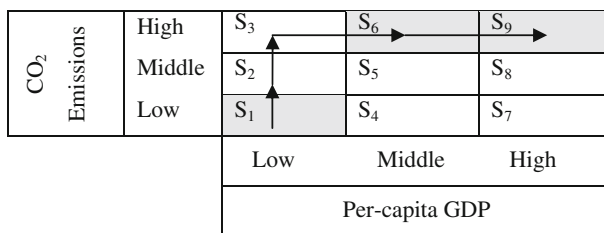
an attractor. Indeed, according to the ergodic distribution estimates, S₁ takes a relatively high value in the steady state (20.1 %).

Moving to the second block (middle-income countries), we observe that transition economies with low pollution have a 6.4 % probability of regressing to low GDP and emission levels and a 9.1 % probability of increasing their pollution levels during one period. Middle-income countries with middle-pollution levels (S₅), have the same probability (1.2 %) of moving up or down the GDP ladder while remaining at the same pollution level (S₂ and S₈). In addition, they exhibit a 4 % probability of becoming more polluted versus a 2.3 % probability of reducing their pollution levels. Finally, even middle-income/high-pollution countries (S₆) tend to remain in the initial state, with only a 4.5 % probability of moving into the highest GDP class (S₉) and a 1 % chance to move into the lowest GDP class (S₃). For these countries (S₆), the probability of reducing emissions is approximately 3 %. According to the estimated ergodic distribution, at the end of the transition process, most middle-income countries will be characterized by high emission levels (14.9 %). Overall, in this group there is a tendency towards middle and high pollution (24.8 %) compared to low pollution (2.3 %).

The last block corresponds to countries with high income. In this class, the highest mobility is observed in the less polluted countries (57.1 %), but this is only because the number of countries in this class is limited. Countries with high income and middle-level pollution (S₈) have an 8.4 % probability of falling into the middle-income class and an 11.7 % probability of experiencing an increase in emission levels. However, countries that move from the high-income class to the middle-income class tend to reduce their emissions (6.7 % of these countries move to state S₅, whereas only 1.7 % of them move to state S₆). Highly polluted and rich countries (S₉), typically the most advanced ones, have more than 98 % probability of remaining in their initial situation, with 1.5 % probability of experiencing a significant reduction in their income level.

The value of the Q-statistic defined in Sect. 4.3 is 48.76. Therefore, considering a χ^2 distribution with 72 degrees of freedom, we do not reject the null that the transition matrix is stationary at a significance level $\alpha = 5\%$. This means that our chain is stationary and the assumption we made at the beginning of the analysis was correct.

The tendency for high-income countries to grow and pollute heavily is confirmed by the ergodic distribution, with 46.2 % of countries in the high-income/high-pollution class. Unfortunately, their growth also implies serious environmental damages. In fact, from the ergodic distribution, we can observe that S₁, S₆ and S₉ are final attractors. However, the presence of

**Fig. 3** Typical development path**Table 4** Distributions over the state-space of Table 1

	S ₁	S ₂	S ₃	S ₄	S ₅	S ₆	S ₇	S ₈	S ₉
1970	0.365	0.032	0.008	0.111	0.111	0.071	0.016	0.024	0.262
2006	0.325	0.071	0.024	0.016	0.119	0.119	0.000	0.008	0.317
Ergodic	0.201	0.035	0.022	0.023	0.099	0.149	0.000	0.008	0.462

three attractors is not in contrast with the existence of a twin-peaked distribution (see Fig. 2), when two of them are contiguous (S₆ and S₉).

Notice that studies based only on GDP or CO₂ dynamics cannot make inferences about vertical or horizontal movements, respectively, whereas our approach allows us to distinguish between these movements as well as shed some light on their sequence. In particular, comparing the transition probabilities between states, we can roughly identify the most likely development paths. Figure 3 shows the path that a typical country tends to have during its development process, if it does not remain in its initial (persistent) state. Therefore, if a country begins in a low-income/low-pollution situation (S₁), it is more likely that it will experience first an increase in pollution, moving to state S₂ and then to S₃ (than moving to S₄). Once a country is characterized by low income/high pollution (S₃), it is most likely to arrive at a state such as S₆ (middle income/high pollution) or S₉ (high income/high pollution). Overall, an initially poor and low-polluting country will possibly experience, initially, a rise in its emission levels and, subsequently, will arrive at the high- (or middle-) income class.

Table 4 reports the cross-country distribution of the first and last years of our sample, along with the ergodic distribution, in terms of the states defined above (see Sect. 4.1). We can identify two main long-run equilibria: (i) low income/low pollution (S₁) and, (ii) high income/high pollution (S₉). The ergodic distribution shows that approximate 63 and 22 % of the world's countries will be characterized by high and low pollution levels, respectively, if our consumption-production habits remain the same. Consequently, our estimations reveal a strong positive relationship between income and CO₂ emissions in the long run.

Moreover, Table 4 indicates how far countries are from the stationary state. The convergence process seems to be quite slow. For instance, the fraction of countries with high income/high pollution levels has increased by 5 % in 36 years, and the remaining distance from the steady state is approximately 15 %.

By considering the state-space in Table 1, Table 5 shows the composition of each GDP class with regard to CO₂ levels in the stationary state. The majority of poor countries (78 %) will be characterized by low pollution, 13.5 % by middle pollution and only 8.5 % by high pollution. Moving to middle-income countries, 54.8 % of these will be characterized by high pollution, 36.7 % by middle pollution and 8.5 % by low pollution. Finally, 98.2 % of rich countries will be highly polluting countries. These findings confirm the presence of a strong positive relationship between income and emissions in the long run. Relating our findings to

Table 5 Ergodic distribution of CO₂ emissions for each GDP class

	Low pollution	Middle pollution	High pollution
Low income	0.780	0.135	0.085
Middle income	0.085	0.367	0.548
High income	0.001	0.018	0.982

Table 6 Mobility indices

Indices	Values	Variance
I_1	0.033	7.01e−06
I_2	0.011	8.48e−07
I_3	0.111	6.33e−05
I_4	0.042	2.14e−05

previous research on carbon dioxide emissions, we confirm the lack of evidence in favor of the EKC hypothesis (see, among others, [Stern 2004](#); [Acaravci and Ozturk 2010](#)). That is, at least for CO₂ emissions, our results do not support the EKC hypothesis as a long-run phenomenon.²³ This can be attributed to a number of reasons ([Stern 2004](#)). On the one hand, the scale effect implies that income increases result in higher pollution for a given production structure and state of technology. On the other hand, an output mix oriented towards less-polluting sectors (e.g., services and light industries) and an input mix replacing more environmentally harmful inputs with less harmful ones tends to reduce emissions. Furthermore, higher productivity and emission-reducing innovations in production processes work in the same direction. Our analysis implies that the scale effect outweighs the remaining effects.

5.2 Additional Results

As for the speed of convergence, the asymptotic half-life is $h = 79$ periods, which implies a very slow convergence process. When h is so high, it becomes merely a signal of low mobility across states instead of a real measure of convergence. Indeed, we cannot assume that transition dynamics will remain the same for almost 160 years, the time required for complete convergence to the ergodic distribution. [Fiaschi and Lavezzi \(2003\)](#) find an asymptotic half-life of approximately 49 periods. Therefore, the divergent dynamics found by [Aldy \(2006\)](#) for CO₂ emissions probably slows down the convergence process.

Table 6 reports the mobility measures previously described (see Sect. 4.4). As we can observe, these results confirm our finding on the speed of convergence. That is, transitions are not fast.

Indices I_2 and I_4 give a higher weight to transitions between distant states to stress the importance of very significant changes in income-pollution status. When we give a higher weight to transitions between distant states, mobility indices decrease in size, implying that significant jumps are not frequent.

Additionally, we conduct some inferential tests on the transition matrix.²⁴ We do so to further explore the income-pollution relationship and shed light on country dynamics with respect to their GDP/CO₂ classes. According to Table 11 (available in “Appendix D”),

²³ Similarly, [Hettige et al. \(2000\)](#) reject the EKC hypothesis for industrial water pollution.

²⁴ A detailed description of our inferential analyses is given in “Appendix D”.

we reject the hypothesis that all GDP classes are equally likely to ultimately have high pollution if they begin from a state of middle-level pollution (Hypothesis (a)). In other words, the income class of a country does matter regarding its movement from the middle- to high-pollution class. In addition, we do not reject the hypothesis that highly polluting countries have the same probability of experiencing high pollution in the future, independently of their income class (Hypothesis (b)). Therefore, high pollution is equally persistent in all GDP classes. We also reject the hypotheses that the probability of countries remaining highly polluting is the same as the probability of falling into the other pollution classes for all GDP groups (Hypotheses (c₁) and (c₂)).

At the same time, according to Table 12 (available in “Appendix D”), for both middle- and high-income countries, we cannot reject the hypothesis that the probability of experiencing middle-level pollution is the same as the probability of high pollution ((Hypothesis (d)). The remaining probabilities of falling into a high-pollution state differ from the probabilities of falling into low or middle-pollution states, beginning from any initial state. Finally, we reject the hypothesis that for each GDP class, the probability of falling into a high-pollution state is the same as the probability of falling into the middle or the low-pollution state ((Hypothesis (e)).

5.3 Robustness Checks

5.3.1 Transition Matrix

Given the problems that might arise from the discretization of the state-space, we use a stochastic kernel technique to represent the CO₂ dynamics for each GDP class in a continuous state-space. Our results are reported in Figs. 4, 5 and 6. These figures correspond to the three blocks we discussed above when the state-space is continuous. On the vertical axis, we have the current value of CO₂ emissions, and on the horizontal axis, we have the estimated value for the next period. Therefore, these figures represent a phase diagram for CO₂ emissions. This means that on the 45-degree line, we find all countries that will remain at exactly the same emission level after one period. Above the 45-degree line, there are countries that will exhibit a reduction in pollution, while below, we have countries that will experience an increase in CO₂ emissions. As illustrated in Fig. 4, a high proportion of countries lies on the 45-degree line, meaning that they tend to remain at almost the same emission level after one period. Nonetheless, all of the countries that lie above the 45-degree line will move back to a lower pollution situation. Figure 4 shows that low-income countries have a relatively high volatility in terms of CO₂ levels; however, their emissions are always below 10 % of the world average. Relative to Figs. 4, 5 and 6 show that middle- and high-income countries exhibit a more homogeneous and stable behavior in terms of carbon dioxide dynamics. Figures 4, 5 and 6 are consistent with the transition matrix presented in Table 3. Moreover, because our contour plots never overlap, we can conclude that the thresholds identified for the sample splitting are sensible.

5.3.2 Ergodic Distribution

As a robustness check, Table 7 reports additional ergodic distributions obtained by changing the sample and the definition of the state-space. In this way, we can compare our previous results (Table 4) with the new results. The first row of the table contains the ergodic distribution derived by the new sample, which consists of 95 countries for the 1960–2006 period. As previously, we can identify two main equilibria, namely, low income/low pollution and

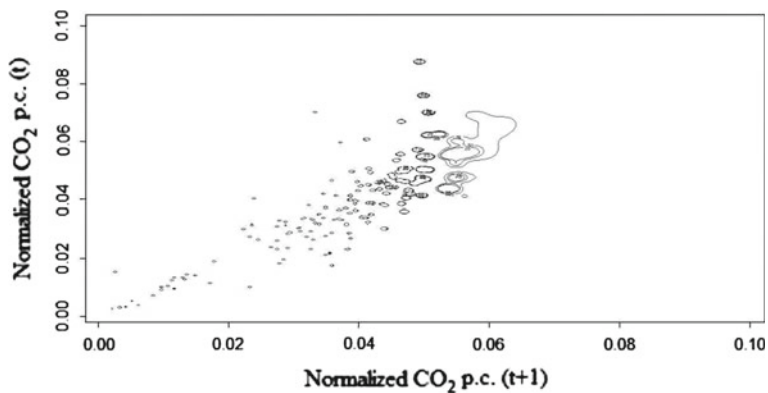


Fig. 4 Stochastic kernel for low-income countries

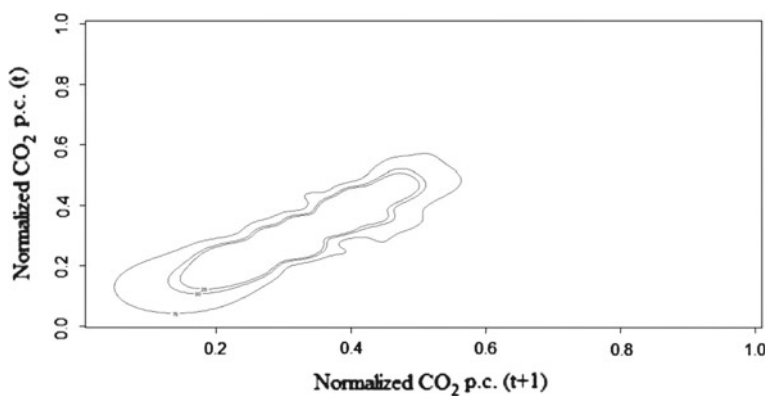


Fig. 5 Stochastic kernel for middle-income countries

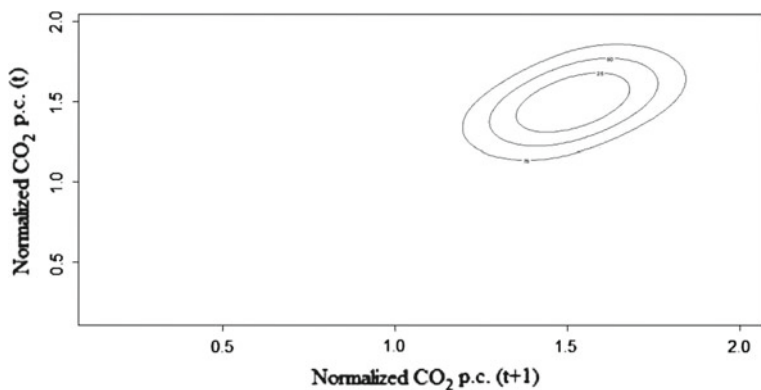


Fig. 6 Stochastic kernel for high-income countries

Table 7 Robustness check

	S ₁	S ₂	S ₃	S ₄	S ₅	S ₆	S ₇	S ₈	S ₉
Ergodic distribution from sample (1960–2006)									
(1960–2006)	0.279	0.028	0.002	0.069	0.117	0.097	0.003	0.016	0.388
Ergodic distributions over the state-space of Table 2									
(1970–2006)	0.111	0.025	0.025	0.009	0.085	0.189	0.000	0.006	0.549
(1960–2006)	0.142	0.016	0.001	0.028	0.114	0.145	0.001	0.019	0.534

high income/high pollution. Furthermore, this row confirms the existence of strong polarization of countries in the CO₂ dimension.

The last two rows of Table 7 provide another robustness check for our analysis. We re-estimate the ergodic distributions of our two samples by using the state-space defined in Table 2 (see Sect. 4.1). These two distributions are similar with respect to the ergodic distributions based on the state-space defined in Table 1, although we have changed the state-space definition, lowering the thresholds in Table 2.

6 Conclusions

By taking an agnostic point of view on *equilibrium multiplicity* vs. *uniqueness*, this paper studies jointly CO₂ and income dynamics. Using a Markov-chain approach with a bivariate state-space, our findings do not support theoretical models predicting that a country can be caught simultaneously in a poverty and environment trap. A strong, long-run, positive relationship between income and carbon dioxide emissions emerges. Regarding transitional dynamics, countries are unlikely to change their income-pollution status. This implies a very slow speed of convergence towards the steady state, which means that carbon emissions might slow down the convergence process. Nevertheless, a country that leaves the low-income/low-pollution situation is most likely to become “dirtier” before getting richer. Overall, the development level of a country affects its CO₂ dynamics. Finally, according to our analysis, in the income-CO₂ space, the EKC could only be a transitory phenomenon.

In general, our findings are compatible with three supplementary hypotheses: governments do not invest enough in air quality maintenance, effective emission-abatement technologies do not exist, and consumption behavior enhances CO₂ levels. This underlines the urgent need for systematic and intense coordinated efforts of the public and private sectors towards less-polluting production technology. Also, consumer behavior should be altered through information provisions as well as economic incentives to reduce pollution. Indeed, because the world’s population is expected to increase dramatically in the coming decades, per-capita carbon dioxide emissions should be reduced substantially to avoid catastrophic consequences for human development.

These results should be taken into account in the context of international negotiations on climate change, where the main issue remains how to share the burden of CO₂ emissions targets between developed and developing economies. At the moment, many developing countries do not have bargaining power to negotiate for licenses because they have neither the financial means to buy licenses nor influence on international markets. However, in terms of environmental sustainability, this paper adds to previous arguments such as *historical responsibility* and *fairness* an even stronger argument: the concept of *future responsibility*.

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Appendices

Appendix A: Estimating Density Functions

Instead of providing simple histograms—with the well known discretization problems— we replace them with smooth kernel functions centered over each observation. In particular, by using a nonparametric method provided by [Bowman and Azzalini \(1997\)](#), we estimate the density function $f(C_t, Y_t)$, where C_t is the relative per capita pollution level at time t and Y_t is the relative per capita GDP at time t . Notice that we do not impose any restriction on the shape of $f(., .)$, which is at least a priori unknown. The kernel estimator can be written as follows:

$$\hat{f}(C_t, Y_t) = \frac{1}{N} \sum_{i=1}^N k(C_t - C_{it}; h_C) k(Y_t - Y_{it}; h_Y),$$

where, (C_t, Y_t) represents the point at which the density is estimated, $\{C_{it}, Y_{it}; i = 1, \dots, N\}$ denote the data and (h_C, h_Y) are two joint smoothing parameters (or bandwidths). Alternatively, we could use a bivariate kernel with correlated components, but a product of univariate components is usually sufficient, especially for descriptive statistics. Functions $k(., .)$ give weights that decrease monotonically as the difference between central points and observations increases. In particular, two univariate normal kernels, one for each explanatory variable, are used and h is obtained by minimizing the mean integrated squared error (MISE). The choice of the optimal smoothing parameter is relevant because, there is a trade-off between bias and variance of our estimates. In fact, whereas variance is decreasing in h , bias is increasing in h .

Appendix B: Country Lists and Summary Statistics

See Tables 8, 9, 10.

Table 8 1970–2006

Afghanistan	Gabon	Niger
Albania	Gambia, The	Nigeria
Algeria	Ghana	Norway
Angola	Greece	Pakistan
Antigua and Barbuda	Grenada	Panama
Argentina	Guatemala	Papua New Guinea
Australia	Guinea	Paraguay
Austria	Guinea-Bissau	Peru
Barbados	Guyana	Philippines
Belgium	Haiti	Poland

Table 8 continued

Belize	Honduras	Portugal
Benin	Hungary	Qatar
Bolivia	Iceland	Romania
Brazil	India	Rwanda
Brunei Darussalam	Indonesia	Sao Tome and Principe
Bulgaria	Iran, Islamic Rep.	Saudi Arabia
Burkina Faso	Iraq	Senegal
Cambodia	Ireland	Sierra Leone
Cameroon	Israel	Singapore
Canada	Italy	Solomon Islands
Central African Republic	Jamaica	South Africa
Chad	Japan	Spain
Chile	Jordan	Sri Lanka
China	Kenya	St. Lucia
Colombia	Korea, Rep.	St. Vincent & Grenadines
Congo, Dem. Rep.	Lao PDR	Sudan
Congo, Rep.	Lebanon	Suriname
Costa Rica	Liberia	Sweden
Cote d'Ivoire	Libya	Switzerland
Cuba	Luxembourg	Thailand
Cyprus	Macao SAR, China	Togo
Denmark	Madagascar	Trinidad and Tobago
Djibouti	Mali	Tunisia
Dominica	Mauritania	Turkey
Dominican Republic	Mauritius	Uganda
Ecuador	Mexico	United Arab Emirates
Egypt, Arab Rep.	Mongolia	United Kingdom
El Salvador	Morocco	United States
Equatorial Guinea	Mozambique	Uruguay
Ethiopia	Nepal	Venezuela, RB
Fiji	The Netherlands	
Finland	New Zealand	
France	Nicaragua	

Table 9 1960–2006

Algeria	Indonesia	Togo
Argentina	Iran	Trinidad & Tobago
Australia	Ireland	Tunisia
Austria	Israel	Turkey
Barbados	Italy	Uganda
Belgium	Jamaica	United Kingdom
Benin	Japan	United States
Bolivia	Jordan	Uruguay

Table 9 continued

Brazil	Kenya	Venezuela
Burkina Faso	Korea, Republic	
Cameroon	Luxembourg	
Canada	Madagascar	
Central African Republic	Mali	
Chad	Mauritania	
Chile	Mauritius	
China	Mexico	
Colombia	Morocco	
Congo, Dem. Rep.	Mozambique	
Congo, Republic of	Nepal	
Costa Rica	The Netherlands	
Cote d'Ivoire	New Zealand	
Cyprus	Nicaragua	
Denmark	Niger	
Dominican Republic	Nigeria	
Ecuador	Norway	
Egypt	Pakistan	
El Salvador	Panama	
Equatorial Guinea	Papua New Guinea	
Ethiopia	Paraguay	
Fiji	Peru	
Finland	Philippines	
France	Portugal	
Gabon	Romania	
Gambia, The	Rwanda	
Ghana	Senegal	
Greece	Sierra Leone	
Guatemala	Singapore	
Guinea	South Africa	
Guinea-Bissau	Spain	
Haiti	Sri Lanka	
Honduras	Sweden	
Iceland	Switzerland	
India	Thailand	

Table 10 Summary statistics

	GDP 1970	GDP 2006	CO ₂ 1970	CO ₂ 2006
N. Obs.	126	126	126	126
Min.	336.482	370.211	0.016	0.025
First quartile	1785.081	2509.799	0.316	0.543
Median	3669.131	7452.452	0.988	2.201
Third quartile	9935.604	21349.795	5.570	6.657
Max.	78724.648	76227.547	67.929	46.131

Table 10 continued

	GDP 1970	GDP 2006	CO ₂ 1970	CO ₂ 2006
Mean	7280.113	13637.600	4.750	4.783
Kurtosis	24.190	3.192	22.906	13.336
Asymmetry	4.223	1.705	4.604	3.102
Std. Dev.	10066.841	15102.786	10.952	6.710
Shapiro–Wilk test	0.584*	0.791*	0.424*	0.676*
Anderson–Darling A ²	12,468*	8,667*	22,835*	10,484*

*We reject the normality hypothesis at the 5 % level of confidence

Appendix C: Definition of the State-Space

In Fig. 1, we saw that in 1970 the joint distribution between GDP and CO₂ emissions were unimodal and extremely skewed; vice versa, Fig. 2 showed the emergence of a second peak. The next figure puts together the level curves of the two density functions (the solid curves denote the initial distribution, whereas the dotted ones represent the final distribution) (Fig. 7).

As we can see, the initial peak acts as a separating equilibrium, with a fraction of countries converging towards lower combinations of GDP and CO₂ and another fraction towards higher combinations. That is, the initial peak seems to define two different basins of attrac-

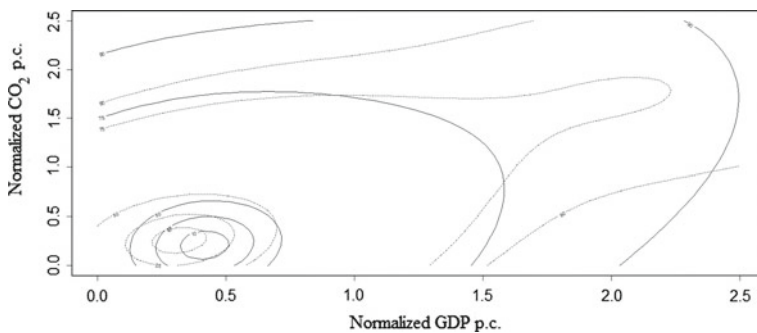


Fig. 7 Contour plots of Fig. 1 and 2

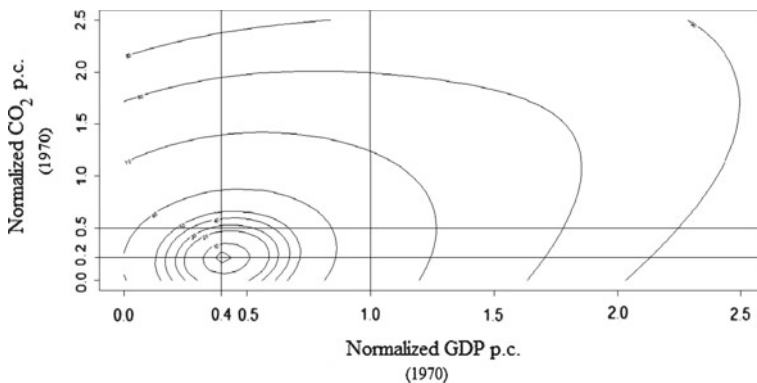


Fig. 8 State-space division CO₂ emissions and income dynamics:

tion. Therefore, the initial peak is naturally chosen as the first critical point to split the sample. Subsequently, we used as a second threshold for CO₂ emissions the sustainable threshold identified with half the current emissions. Obviously, over time technological progress as well as greener behaviors could lead to a sustainable threshold higher than this value, but unfortunately, looking at the average emissions in 2006 reported in Table 10 (Summary statistics) it does not seem to be the current tendency. Finally as second threshold for GDP we took the world average, since it allowed us to have almost equal size subsamples. This is due to the fact that the income distribution is extremely asymmetric. The next figure shows the graphical technique used to split our state-space (Fig. 8).

Appendix D: Hypothesis Testing

This section presents the inferential analysis we conduct on the transition matrix. For regular Markov chains, the difference between the elements of the transition probabilities after a number of transitions tends to zero. In other words, all elements of all columns tend towards a common limit as the number of transitions increases. At the same time, [Stuart and Ord \(1994\)](#) show that each row of the transition matrix converges to a N-variate normal distribution, that is, $\sqrt{N_i}(\hat{p}_{ij} - p_{ij})$ asymptotically converges towards a normal distribution with zero mean and variance $p_{ij}(1 - p_{ij})$. We can use this result to conduct inference on the elements of the transition matrix. By using a standard normal distribution, we can compare elements of the same row as follows:

$$s^2 = \frac{N_i(\hat{p}_{ij} - \hat{p}_{iq})^2}{\hat{\sigma}_{ij}^2 + \hat{\sigma}_{iq}^2 + 2N_i\hat{p}_{ij}\hat{p}_{iq}}$$

where s^2 is the sample variance, N_i is the total number of observations in state i , j and q are two general states at the end of the period, $\hat{p}$ are the estimated probabilities and $\hat{\sigma}^2$ are the estimated variances.

Similarly, we can compare elements of different rows (i and z)

$$s^2 = \frac{N_i N_z (\hat{p}_{ij} - \hat{p}_{zq})^2}{N_z \hat{\sigma}_{ij}^2 + N_i \hat{\sigma}_{zq}^2}.$$

Given these results, we test specific hypotheses concerning the transitional dynamics. Under the assumption that rows of the transition matrix are independent, we run the following p-value tests. Since the transition matrix depends on the state-space definition, we decided to consider a p value of less than 0.10 as significant.

Tests

- a. *Is pollution equally increasing among GDP classes?* We test whether low and high-income countries have the same probability of middle-income countries to move from the middle to high-pollution status.

$$H_0 : p_{53} + p_{56} + p_{59} = p_{23} + p_{26} + p_{29}$$

$$H_0 : p_{53} + p_{56} + p_{59} = p_{83} + p_{86} + p_{89}$$

- b. *Is high pollution equally persistent among GDP classes?* We test if the probability of occurrence of high pollution is the same across GDP classes. To do this, we compare

Table 11 Tests of hypotheses a, b and c

Test hypothesis	Low income <i>p</i> value	H ₀	Middle income <i>p</i> value	H ₀	High income <i>p</i> value	H ₀
a	0.4229	R	—	—	1	R
b	0.0163	NR*	—	—	0.0163	NR*
c ₁	0.995	R	1	R	1	R
c ₂	0.928	R	0.921	R	0.954	R

p values measure the probability that H₀ is false. Significance levels: * 2 %, *R* reject, *NR* not reject; c₁ probability to stay in high-pollution class compared with the probability to move in low-pollution class, c₂ probability to stay in high-pollution class compared with the probability to move in middle-pollution class

Table 12 Tests of hypotheses d and e

Income	Test d	<i>p</i> value	H ₀	Test e	<i>p</i> value	H ₀
Low	Low versus high pollution	1	R	S ₃ versus S ₁	1	R
Low	Middle versus high pollution	0.7528	R	S ₃ versus S ₂	1	R
Middle	Low versus high pollution	1	R	S ₆ versus S ₄	1	R
Middle	Middle versus high pollution	0	NR	S ₆ versus S ₅	1	R
High	Low versus high pollution	0.989	R	S ₉ versus S ₇	0.9989	R
High	Middle versus high pollution	0	NR	S ₉ versus S ₈	0.6081	R

p values measure the probability that H₀ is false. Significance levels: *** 2 %, ** 5 %, * 10 %. *R* reject, *NR* not reject

middle-income countries with low and high-income countries respectively with regard to pollution.

$$H_0 : p_{63} + p_{66} + p_{69} = p_{33} + p_{36} + p_{39}$$

$$H_0 : p_{63} + p_{66} + p_{69} = p_{93} + p_{96} + p_{99}$$

- c. *Does a country tend to remain in the high-pollution class?* We investigate whether high pollution is persistent in each GDP class. Denoting with *j*, *z* and *i* a low-, middle- and high-pollution state, respectively, we conduct the following tests:

$$H_0 : p_{ii} = p_{ij} (c_1) \text{ and } H_0 : p_{ii} = p_{iz} (c_2)$$

- d. *Are transition probabilities symmetric with respect to pollution?* Starting from any given state, we want to know if countries have the same probability of going into a high-pollution class compared with low and middle-pollution classes. Then, for any GDP class we test the following hypotheses:

$$H_0 : p_{.,3} + p_{.,6} + p_{.,9} = p_{.,1} + p_{.,4} + p_{.,7}$$

$$H_0 : p_{.,3} + p_{.,6} + p_{.,9} = p_{.,2} + p_{.,5} + p_{.,8}$$

- e. *Is the probability to fall into a high-pollution class equal to the probability to fall into any other pollution class?*

Starting from any initial state, we want to know if countries have the same probability of moving into a high-pollution state relative to low and middle pollution states, given a final GDP class. So, we examine the probability of moving to S_i versus S_j

$$H_0 : p_{,i} = p_{,j}$$

See (Tables 11, 12).

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